MESA COHORT C4 · AI AND ITS APPLICATIONS · PROBLEM 20

Podcast Ecosystem Radar — Components Map

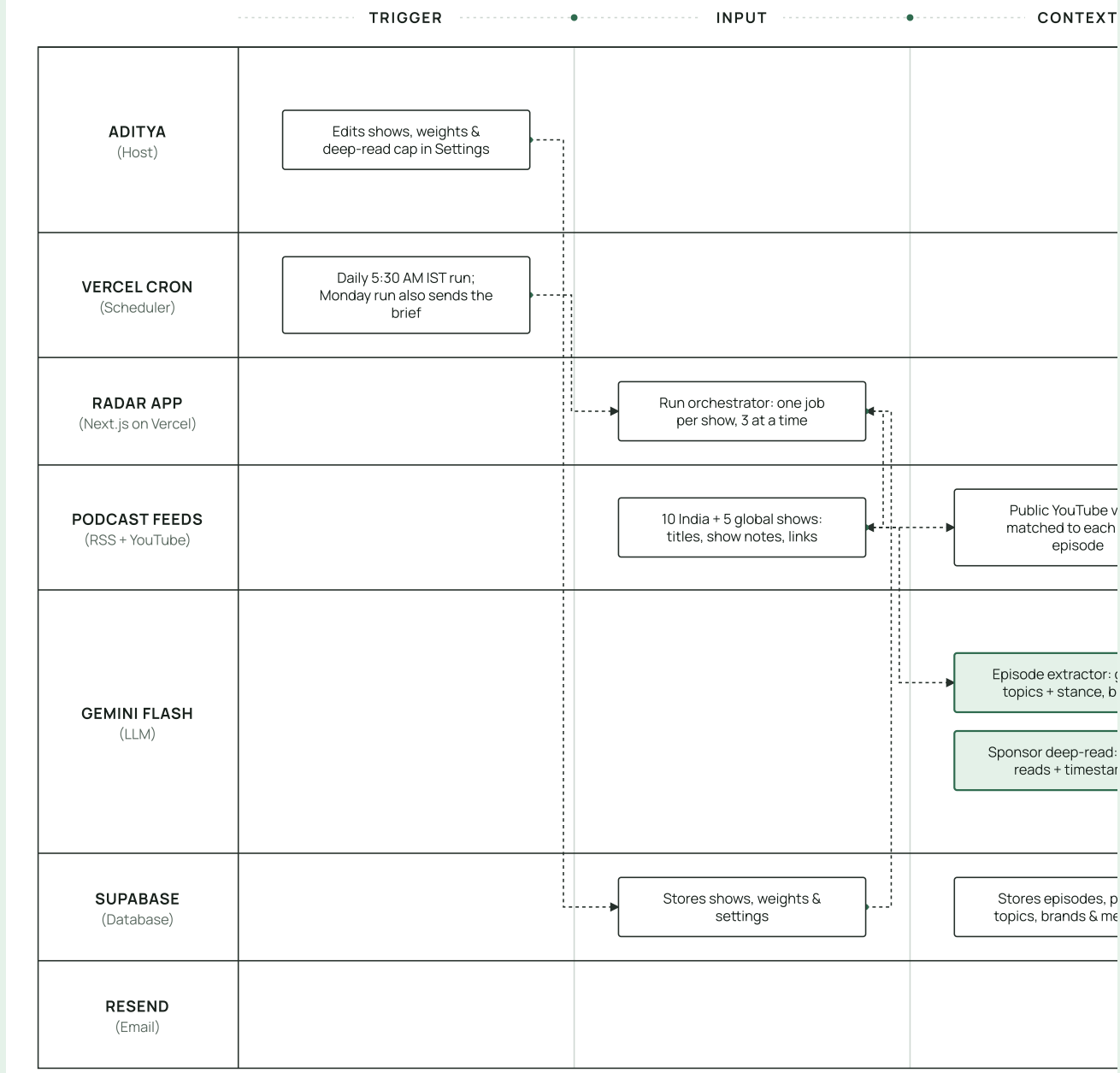
Every Monday before 7:30 AM IST, Aditya gets one sourced brief built from live data across 10 Indian and 5 global podcasts: which founders to book, which topics to ride or counter, and which brands are buying podcast reads. Every AI step runs on Gemini Flash.

LIVE APP
podcast-radar-psi.vercel.app

GITHUB REPOSITORY
github.com/ichhajain-hub/podcast-radar

AI MODEL
gemini-3.8-flash

Components Map



Gemini Flash (5 AI steps) Infrastructure, data and delivery

Dotted arrows show data moving between components

How a Monday run works

1. **Trigger.** Vercel Cron calls `/api/run?mode=weekly` at 5:30 AM IST (daily runs refresh the dashboard the same way, without email).
2. **Read config.** The orchestrator loads the active shows and weights Aditya set in Settings. No code or redeploy needed.

3. **Fetch.** One job per show pulls live RSS or YouTube-channel feeds: the last 30 days for India, 14 days for global. Episodes already stored are skipped.
4. **Extract.** Gemini Flash reads titles and show notes and returns guests, topics with stance, and brands, each with a short evidence quote.
5. **Deep-read.** For India episodes with a public YouTube video, Gemini Flash watches it at low resolution and lists spoken sponsor reads with timestamps.
6. **Resolve.** Gemini Flash merges name variants of people, themes and brands. Duplicates are linked, never deleted.
7. **Score.** `score(from, to, weights)` applies the gates and weights and tags each item New, Rising or Repeat against last week.
8. **Find links.** Gemini Flash with Google Search grounding finds LinkedIn and company sites for shortlisted guests, and keeps only URLs from search results.
9. **Write and deliver.** Gemini Flash writes the "why now" lines from each item's own sources (fact-checked in code). The brief is stored, emailed through Resend and shown on the dashboard.

Component inventory

#	COMPONENT	TOOL	ROLE
1	Settings dashboard	`Next.js + Supabase`	Password-protected page to add, edit and switch shows (India / global / own), edit weights, the guest gate and the video deep-read cap; shows estimated spend
2	Scheduler	`Vercel Cron`	Daily 5:30 AM IST refresh; Monday 5:30 AM IST run also sends the brief (lands by ~6:35)
3	Run orchestrator	`Next.js API route`	One ingest job per show, 3 at a time; up to 2 video deep-reads in parallel; a failed show is logged and named in the brief
4	Feed fetcher	`rss-parser`	Live RSS and YouTube channel feeds; links each India episode to its public YouTube video
5	Episode extractor	`Gemini Flash`	Structured JSON (validated with zod): guests, topics + stance, brands + paid/organic, each with an evidence quote

#	COMPONENT	TOOL	ROLE
6	Sponsor deep-read	`Gemini Flash`	YouTube URL at low media resolution; spoken sponsor reads with timestamps; minute cap set in Settings
7	Entity resolver	`Gemini Flash`	Maps new names onto canonical people, topics and brands; groups topic themes
8	Link finder	`Gemini Flash + Search`	LinkedIn and company site from grounded search results only; otherwise a "Search on LinkedIn" link
9	Store	`Supabase Postgres`	Permanent history: shows, weights, settings, episodes, entities, mentions, briefs, runs (with token usage)
10	Scoring engine	`TypeScript score()`	Gates, weighted scores out of 100, New / Rising / Repeat; one function for daily, weekly and history views
11	Brief writer	`Gemini Flash`	"Why now" lines and India angles using only each item's sources; rejected and retried if it names anything else
12	Delivery	`Resend + dashboard`	Monday email; dashboard with date filter (As sent / Explore), archive and run log

How the problem statement is met

REQUIREMENT	HOW IT IS MET	COMPONENTS
Live external data	Real RSS feeds and public YouTube videos fetched every morning; 8 weeks backfilled	2, 4, 6
Gemini Flash for all AI; no Claude API	Extraction, deep-read, name merging, link finding and brief writing all call Gemini Flash; an automated test fails if the other vendor's SDK or API appears in the repo	5, 6, 7, 8, 11
Deployed on Vercel	App, API routes and crons on Vercel; public URL and public GitHub repo	2, 3, 12

REQUIREMENT	HOW IT IS MET	COMPONENTS
10 shows, configurable without code	Settings page with add / edit / switch-off and a live feed check; changes apply on the next run	1
Guests on multiple shows in the same month	30-day cross-show count; a "Doing media rounds" block lists every guest on 2+ India shows, and "Book this week" ranks founders who are ahead of the curve	7, 10
Topics across shows	Weekly show count, acceleration vs the 4-week average, ride / counter tag, plus global signals not yet covered in India	10
Brands mentioned or promoted	Show notes plus spoken reads, tagged paid or organic, with timestamped links	5, 6, 10
Brief every Monday morning	Monday 5:30 AM IST cron; email via Resend, arriving before 7:30 AM IST	2, 12